

Scalability & Future Plan

Date	03 October 2026
Team ID	SWTID-2026-6945
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

This document explains how ComicCraft can grow to serve more users and how it will evolve.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Comic generation runs inside the web request and image inference is serialised	Users wait minutes; concurrent users queue for the GPU	High
2	No login, rate limiting or content moderation	Anyone can consume API quota; risk of unsuitable prompts	High
3	Generated files stay on local disk with no clean-up and no database	Disk grows over time; comics cannot be revisited later	Medium
4	A single Stable Diffusion model per process; no character consistency across panels	Limited visual continuity	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	One Uvicorn process, work done in request threads	Background job queue (Celery/RQ + Redis) with progress polling; several workers behind Nginx or Cloud Run
2	Data Storage	Local folders static/panels and static/exports	Cloud object storage for images and PDFs, database for users and comics, scheduled clean-up
3	Performance	Sequential images, 25 inference steps, GPU or CPU	GPU inference service, faster/distilled image model, parallel panels across GPUs, caching of outlines
4	Security	Secrets in .env, input validation	User authentication, rate limiting, prompt/content moderation, secret manager, HTTPS

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Accounts, saved comic library, regenerate a single panel, edit text	Q4 2026	Users can revisit and refine comics
Phase 3	Consistent characters, voice narration, multi-language stories, content-safety filter	Q1 2027	Higher quality and wider audience
Phase 4	Cloud deployment with GPU, CI/CD, share links, more panel counts and layouts	Q2 2027	Reliable public service that scales